

AI CREDIT RISK PREDICTION SYSTEM

Technical & Business Documentation

Author: Shaqib Khan

1. Project Overview

The AI Credit Risk Prediction System is a full-stack machine learning application designed to assist banks and digital lending platforms in evaluating the probability of loan default for applicants.

The system integrates:

- Machine Learning model for prediction
- Secure backend API
- Authentication system for banks
- Interactive frontend dashboard
- Cloud deployment architecture

It enables institutions to make faster, data-driven, and risk-aware lending decisions.

2. Business Problem

Financial institutions face significant losses due to incorrect loan approvals. Traditional rule-based systems are often insufficient to capture complex borrower risk patterns.

Key challenges:

- Increasing loan volume while minimizing defaults
 - Evaluating applicant risk objectively
 - Ensuring regulatory interpretability
 - Maintaining operational efficiency
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3. Business Objective

- Reduce loan default rates
- Improve quality of loan approvals
- Provide consistent risk scoring

4. Machine Learning Objective

To predict the probability of default for a loan applicant using historical borrower data and categorize applicants into:

- LOW risk
- MEDIUM risk
- HIGH risk

5. Dataset Description

Dataset size:

100,000+ records

Features:

Feature	Description
age	Applicant age
income	Annual income
credit_score	Credit bureau score
employment_type	Employment category
loan_amount	Requested loan
loan_tenure	Loan duration (months)
emi_to_income_ratio	EMI burden
past_default_history	Previous defaults
default_status	Target variable

Dataset characteristics:

- Realistic financial distributions
- Missing values
- Noise
- Outliers

Designed to reflect real-world lending data.

6. Data Preprocessing

- Missing value imputation

- Feature scaling (StandardScaler)
 - Categorical encoding (OneHotEncoder)
 - Feature consistency enforcement using Pipeline
 - Data validation before prediction
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7. Model Development

Models tested:

- Logistic Regression
- Random Forest
- Gradient Boosting

Final model selected:

Logistic Regression

Reasons:

- High interpretability
- Stable performance
- Regulatory friendly
- Lower risk of overfitting
- Transparent probability output

Final performance (approx):

Metric	Value
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ROC-AUC	0.88–0.90
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Accuracy	0.85–0.89
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Precision	Balanced
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Recall	Balanced
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8. Risk Classification Logic

Probability	Risk Category
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< 35%	LOW
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Probability Risk Category

35% – 60% MEDIUM

> 60% HIGH

9. System Architecture

User (Browser)

v

Netlify Frontend (HTML/CSS/JS)

v

FastAPI Backend (Render)

v

ML Model (Pickle)

v

SQLite Database (Users)

10. Backend Design (FastAPI)

Core components:

- /register – Bank registration
- /login – Authentication
- /predict – Risk prediction (protected)
- Token validation middleware
- CORS configuration
- Model loading via pickle

Security:

- bcrypt password hashing
 - Token-based authentication
 - Protected endpoints
 - SQL injection safe queries
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11. Frontend Design

Pages:

- Login page
- Registration page
- Dashboard

Features:

- Secure authentication
 - Input validation
 - API integration
 - Real-time risk visualization
 - Responsive layout
 - Professional banking UI
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12. Deployment

Backend:

- Platform: Render
- Server: Uvicorn
- Environment: Python
- Model loading at startup

Frontend:

- Platform: Netlify
 - Static hosting
 - API calls to Render backend
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13. Authentication Flow

1. Bank registers
2. Password hashed and stored
3. Bank logs in
4. Token issued

5. Token sent with prediction requests
 6. API validates token
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14. Error Handling

- Invalid credentials handling
 - Feature mismatch validation
 - Network failure detection
 - API error responses
 - Input validation on frontend
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15. Key Technical Challenges Solved

- Feature mismatch between training and deployment
 - CORS policy configuration
 - Model version compatibility
 - NaN prediction errors
 - Token validation
 - UI alignment issues
 - Production deployment errors
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16. Project Outcomes

- Fully working ML prediction system
 - Secure authentication
 - Production deployment
 - Business-aligned risk scoring
 - Realistic evaluation metrics
 - Professional UI
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17. Future Enhancements

- SHAP explainability

- PostgreSQL database
 - Admin panel
 - Loan history tracking
 - Model monitoring
 - Cloud storage
 - Multi-bank analytics dashboard
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18. Conclusion

This project demonstrates complete ownership of:

- Data preparation
- Machine learning modeling
- API development
- Authentication systems
- Frontend engineering
- Cloud deployment
- Business problem translation

It represents a production-grade AI application suitable for real financial use cases.